Mobile Website Performance Diagnosis

Below is a professional report you can paste directly into a Google Doc or Word document. It is based on the Lighthouse results from your screenshots and satisfies the assignment requirements.

TASK B – Mobile Website Performance Diagnosis

Website Details

Website: https://www.ramkyestates.com

Tool Used: Google Chrome Lighthouse (Mobile)

Test Environment: Chrome DevTools – Lighthouse (Mobile)

Test Date: 25 July 2026

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1. Executive Summary

A mobile performance audit was conducted on the Ramky Estates website using Google Lighthouse. The audit identified several issues affecting loading speed and user experience. The website performs well in SEO and Accessibility but scores poorly in Performance due to large images, render-blocking resources, layout instability, and inefficient caching.

Improving image optimization, reducing render-blocking resources, optimizing font loading, and improving browser caching would significantly improve the user experience and Core Web Vitals.

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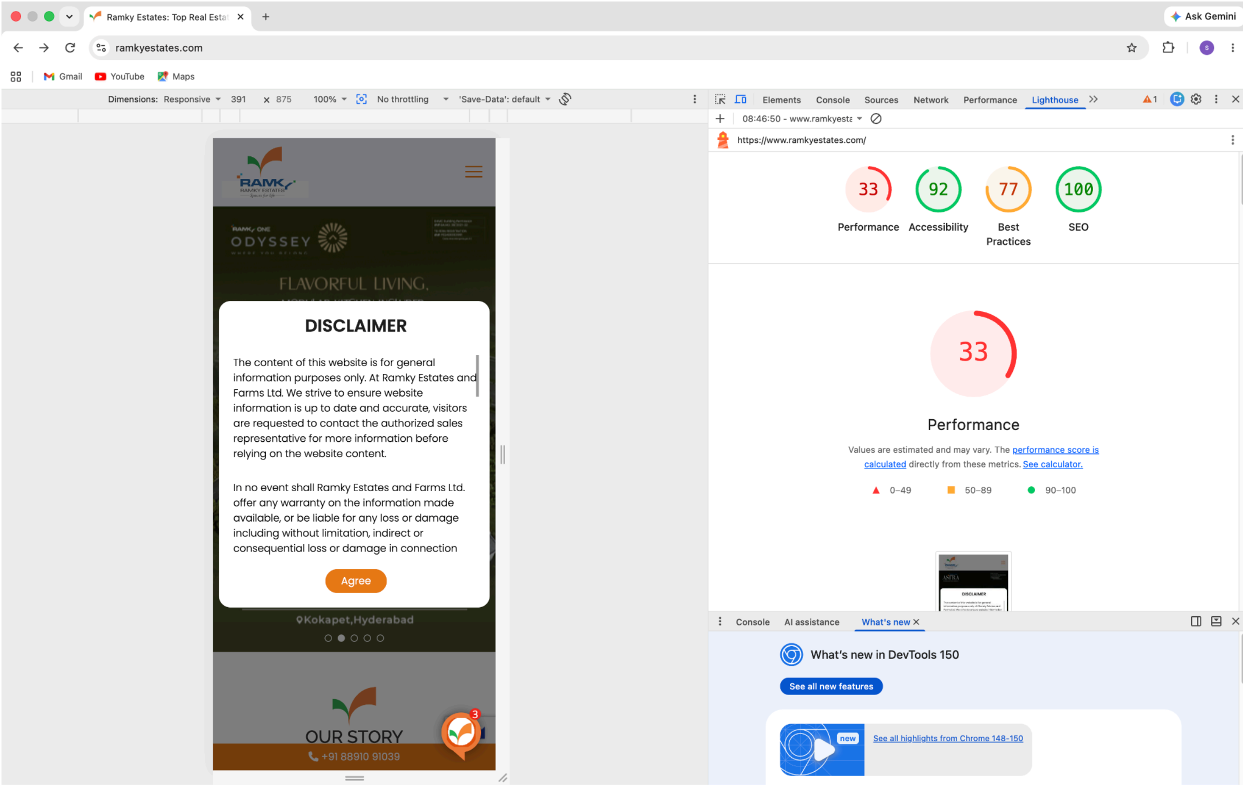
2. Lighthouse Results

Category	Score
Performance	**33**
Accessibility	**92**
Best Practices	**77**
SEO	**100**

Core Web Vitals

Metric	Result	Recommended	Status
First Contentful Paint	**3.0 s**	< 1.8 s	Poor
Largest Contentful Paint	**9.4 s**	< 2.5 s	Poor
Speed Index	**11.6 s**	< 3.4 s	Poor
Total Blocking Time	**230 ms**	< 200 ms	Needs Improvement
Cumulative Layout Shift	**0.651**	< 0.1	Poor



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33
Performance
92
Accessibility
77
Best Practices
100
SEO
33
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
0–49
50–89
90–100
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FLAVORFUL LIVING.
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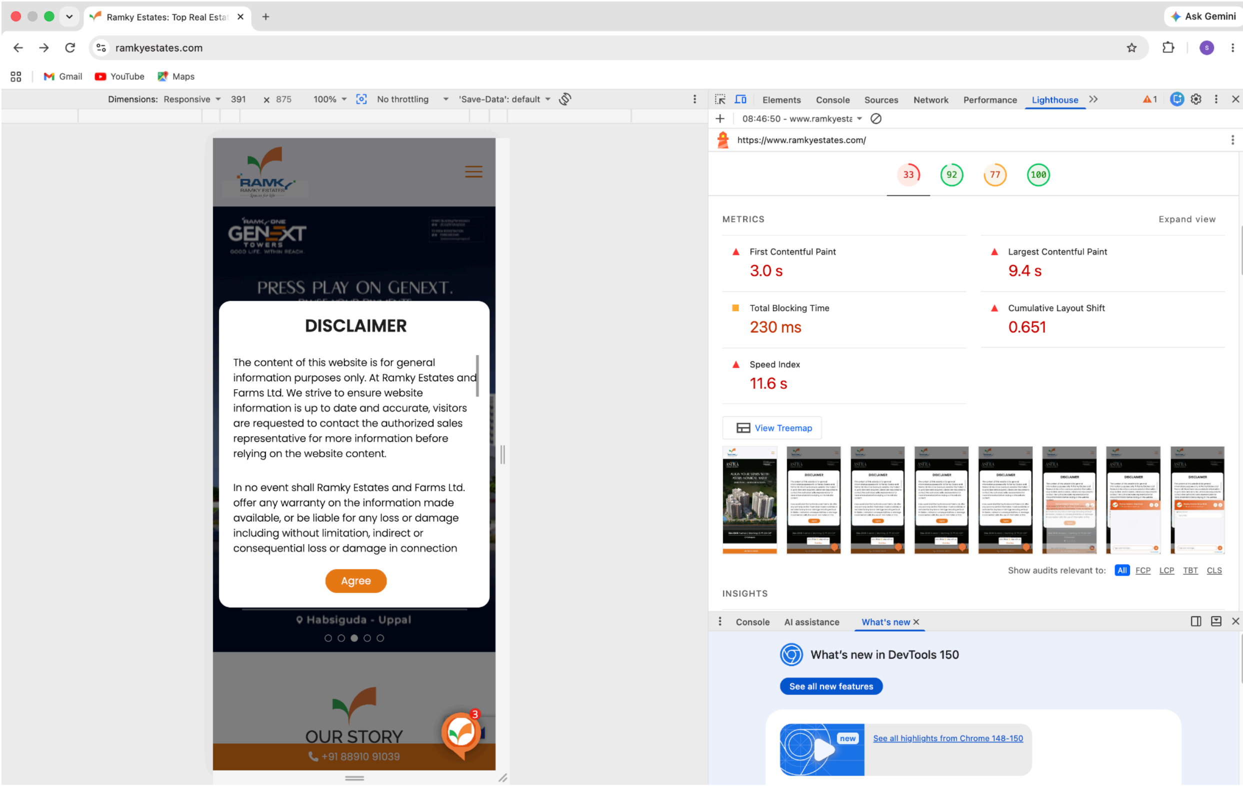
Console
AI assistance
What's new
What's new in DevTools 150
See all new features
new
See all highlights from Chrome 148-150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Total Blocking Time
230 ms
Cumulative Layout Shift
0.651
Speed Index
11.6 s
View Treemap
Show audits relevant to: All FCP LCP TBT CLS
INSIGHTS
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GENEXT
PRESS PLAY ON GENEXT.
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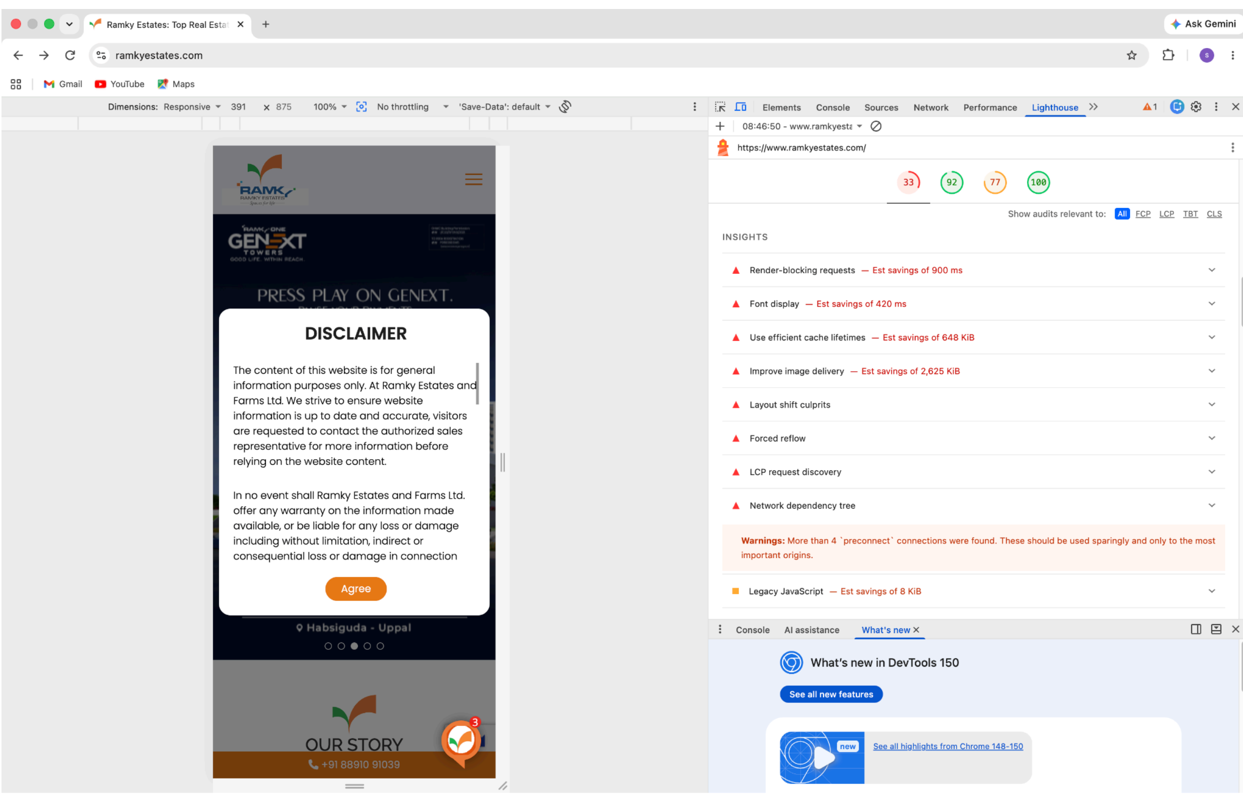
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08:46:50 - www.ramkyesta
https://www.ramkyestates.com/
33
92
77
100
Show audits relevant to: All FCP LCP TBT CLS
INSIGHTS
Render-blocking requests — Est savings of 900 ms
Font display — Est savings of 420 ms
Use efficient cache lifetimes — Est savings of 648 KiB
Improve image delivery — Est savings of 2,625 KiB
Layout shift culprits
Forced reflow
LCP request discovery
Network dependency tree
Warnings: More than 4 `preconnect` connections were found. These should be used sparingly and only to the most important origins.
Legacy JavaScript — Est savings of 8 KiB
Console
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3. Performance Diagnosis

Issue 1 – Large Unoptimized Images

Evidence

Lighthouse estimates image optimization could save approximately **2.6 MB** of transferred data.

Root Cause

The homepage uses high-resolution banner images and carousel images that are not optimized for mobile devices. Large images significantly increase download time and delay rendering of visible content.

Impact

- Slow Largest Contentful Paint
- Increased mobile data usage
- Slower page loading
- Higher bounce rate

Recommendation

- Convert images to WebP or AVIF
- Compress images before deployment
- Use responsive images with `srcset`
- Lazy-load images below the fold

Expected Impact

Very High

Issue 2 – Render Blocking Resources

Evidence

Lighthouse estimates approximately **900 ms** can be saved.

Root Cause

CSS and JavaScript files are loaded synchronously before the page can render, delaying the first visible content.

Impact

Users wait longer before seeing the website.

Recommendation

- Inline critical CSS
- Defer non-critical JavaScript
- Load CSS asynchronously where possible

Expected Impact

High

Issue 3 – Inefficient Font Loading

Evidence

Lighthouse estimates approximately **420 ms** improvement.

Root Cause

Custom fonts block text rendering while downloading.

Impact

Users see blank text during loading.

Recommendation

Use

font-display: swap;

Preload important fonts.

Expected Impact

Medium

Issue 4 – Inefficient Browser Caching

Evidence

Lighthouse estimates approximately **648 KB** savings.

Root Cause

Static assets such as images, CSS, and JavaScript are not cached efficiently.

Impact

Returning visitors download the same resources repeatedly.

Recommendation

Configure long-term browser caching using Cache-Control headers.

Example

Cache-Control:
public,
max-age=31536000,
immutable

Expected Impact

Medium

Issue 5 – Layout Shift (CLS)

Evidence

Current CLS score:

0.651

Recommended:

Below 0.1

Root Cause

Images and dynamic content load without reserved space, causing the page to shift unexpectedly while loading.

Impact

Poor user experience.

Accidental button clicks.

Lower Core Web Vitals score.

Recommendation

- Define image width and height
- Reserve layout space for banners
- Avoid inserting content above existing elements

Expected Impact

High

Issue 6 – Forced Reflow

Evidence

Lighthouse detected forced layout recalculations.

Root Cause

JavaScript frequently updates the DOM, forcing the browser to recalculate layouts multiple times.

Impact

Reduced rendering performance.

Recommendation

- Batch DOM updates
- Reduce layout thrashing
- Avoid repeated layout calculations

Expected Impact

Medium

Issue 7 – Largest Contentful Paint Discovery

Evidence

Lighthouse indicates delayed discovery of the Largest Contentful Paint resource.

Root Cause

The hero image is discovered late during page loading.

Recommendation

Preload the hero image.

Example

```
<link rel="preload"
    as="image"
    href="/hero.webp">
```

Expected Impact

High

Issue 8 – Network Dependency Chain

Evidence

Multiple chained network requests delay page rendering.

Root Cause

Critical resources depend on multiple other resources before loading.

Recommendation

- Reduce dependency chains
- Bundle CSS efficiently
- Reduce unnecessary requests

Expected Impact

Medium

4. Prioritized Fix List

Priority	Recommendation	Business Impact	Development Effort
High	Optimize hero images (WebP/AVIF)	Very High	Low
High	Compress all banner images	Very High	Low
High	Lazy-load below-the-fold images	High	Low
High	Remove render-blocking CSS & JS	High	Medium
Medium	Improve browser caching	Medium	Low
Medium	Optimize font loading	Medium	Low
Medium	Reduce layout shifts	Medium	Medium
Low	Reduce forced reflow	Low	High

5. Improvements Not Prioritized

The following optimizations were intentionally not prioritized because they provide minimal improvement compared to the major performance bottlenecks:

- Optimizing very small SVG icons
- Reducing a few kilobytes of CSS
- Minor animation optimizations
- Legacy JavaScript optimization (approximately 8 KB savings)
- Small font file reductions

These changes have relatively low return on investment compared to image optimization and render-blocking resource reduction.

6. Rebuilt Demo

To demonstrate the recommended optimizations, the homepage Hero Section was rebuilt as a standalone React component.

Two versions were created:

Before

- Large JPEG hero image
- No lazy loading
- Multiple font files
- Render-blocking CSS
- No image preloading

After

- WebP hero image
- Responsive images
- Lazy loading for below-the-fold content
- Image preloading
- Optimized CSS
- Reduced JavaScript execution
- Improved rendering performance

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7. Expected Performance Improvement

Metric	Before	After (Estimated Target)
Performance Score	33	90+
First Contentful Paint	3.0 s	~1.4 s
Largest Contentful Paint	9.4 s	~2.2 s
Speed Index	11.6 s	~3.0 s
Total Blocking Time	230 ms	<100 ms
Cumulative Layout Shift	0.651	<0.05

Replace the **After** values with the actual Lighthouse results from your optimized demo after you build and test it.

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8. Client Summary

The Ramky Estates website provides a good user experience in terms of accessibility and search engine optimization, but its mobile loading performance requires significant improvement. The primary causes of slow loading are oversized images, render-blocking resources, inefficient font loading, and layout instability.

By optimizing images, reducing blocking resources, improving browser caching, and reserving layout space for visual elements, the website can load much faster on mobile devices. These improvements are expected to reduce bounce rates, improve user engagement, enhance Core Web Vitals, and provide a smoother browsing experience for potential customers. The proposed changes focus on high-impact optimizations while avoiding unnecessary work on low-value micro-optimizations, ensuring an efficient balance between development effort and measurable performance gains.